TM1026M Series

Product Specification

Integrated Semiconductor Fingerprint Module

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English translation of the supplied Chinese datasheet

Contents

1. Product Overview

2. Technical Specifications

2.1 Performance Parameters

2.2 Electrical Parameters

3. Product Appearance and Mechanical Dimensions

4. Communication Interface Definition

5. Low-Power Reference Design

6. Fingerprint Module Operating Flow

Appendix 1: Fast Communication / Handshake

Appendix 2: Customer Usage Notes

Revision History

1. Product Overview

The Belsec TM1026M series integrated semiconductor fingerprint module consists primarily of the TS1026M fingerprint sensor, the TA0702 fingerprint chip, and fingerprint-recognition algorithms, all developed using the company's proprietary intellectual property.

The proprietary technology is intended to provide customers with efficient and flexible support for secondary development while meeting customer requirements and avoiding intellectual-property disputes. Chip integration also significantly reduces the module's size. The simple structure and modular design improve product stability and consistency.

The TM1026M series provides a fingerprint-processing platform that can communicate with an external controller (host) over a serial port using the TM1026M integrated-program communication protocol, making secondary development convenient.

2. Technical Specifications

2.1 Performance Parameters

Parameter	Specification
Image resolution	160 × 160 pixels
Image resolution density	508 DPI
Chip package	12 mm × 12 mm × 0.6 mm
Module package	33 mm × 20 mm × 6.6 mm
Matching speed (1:1)	< 4 ms per fingerprint
Startup time	< 140 ms
Image acquisition time	< 150 ms
False Rejection Rate (FRR)	< 1%
False Acceptance Rate (FAR)	< 0.001%
Storage capacity	225 fingerprint records standard; expandable on request
Service life	1,000,000 operations

2.2 Electrical Parameters

Parameter	Minimum	Typical	Maximum	Unit
Touch-sensor supply voltage	2.5	3.3	5.5	V
Fingerprint module supply voltage	2.7	3.3	3.6	V
Operating current	20	30	45	mA
Quiescent current	5	7	10	µA
Operating temperature	–20	—	70	°C
Operating humidity (non-condensing)	40	—	85	% RH
Storage temperature	–40	—	85	°C
Storage humidity (non-condensing)	—	—	85	% RH
ESD, air discharge	—	—	15K	V
ESD, contact discharge	—	—	8K	V

Notes: Operating current refers to the current drawn while the fingerprint module is acquiring an image, such as during enrollment or matching. Quiescent current is specified with the fingerprint supply at 0 V and the touch-sensor supply at 3.3 V.

3. Product Appearance and Mechanical Dimensions

The original datasheet provides top-view and bottom-view illustrations, a six-pin connector orientation, and a mechanical-dimension drawing. The module dimensions are 33 mm × 20 mm × 6.6 mm; the drawing specifies a tolerance of ±0.1 mm.

Figure references in the source: Figure 2.1 — Product appearance; Figure 2.2 — Mechanical dimensions (tolerance ±0.1 mm).

4. Communication Interface Definition

Communication interface: standard UART at TTL logic levels.

Default baud rate: 115200 bps; 1 start bit, 1 stop bit; 3.3 V TTL logic. Connector: MX1.25-6P, horizontal SMT type.

Pin assignment (wire order as shown in Figure 2.1):

Pin	Signal	Description
1	V_TOUCH	3.3 V touch-sensor supply; must remain powered continuously.
2	TOUCH_OUT	Wake-up IRQ output (true = 1, false = 0).
3	VCC	Fingerprint module supply.
4	TX	UART_TX: fingerprint module → MCU.
5	RX	UART_RX: MCU → fingerprint module.
6	GND	Ground.

Notes: The serial interface uses 3.3 V TTL levels; use a level converter when connecting it to a computer. V_TOUCH must remain powered. TOUCH_OUT is the touch output, and its voltage level follows V_TOUCH. By default, logic 1 (V_TOUCH) indicates detection, while logic 0 indicates no detection. While the fingerprint sensor is acquiring an image, TOUCH_OUT indicates detection and does not respond to a finger press. After image acquisition ends, it can respond to finger presses. For example, after an enrollment command is sent, the sensor enters image-acquisition mode and TOUCH_OUT indicates detection.

5. Low-Power Reference Design

Module power consumption can be reduced by switching the fingerprint module's VCC supply on and off. In the reference circuit, VIN is the host's 3.3 V supply, VCC is the fingerprint module supply, and CTRL is a host-MCU I/O control signal. CTRL controls power to the module's VCC rail.

Source figure: Figure 5.1 — 3.3 V power-control circuit.

6. Fingerprint Module Operating Flow

The source describes two approaches for waiting for module readiness: a fixed 150 ms delay in the host MCU, or the fast-start handshake described in Appendix 1.

A. Host MCU begins operating the fingerprint module.

B. Host MCU enables the fingerprint module's VCC supply.

C. Host MCU waits for module initialization (150 ms), or performs the fast-start handshake.

D. Host MCU sends the required command (CMD).

E. Host MCU waits for the command to finish. If another command is needed, repeat the command process.

F. Host MCU disables the fingerprint module's VCC supply.

Appendix 1: Fast Communication / Handshake

The following is the translated explanation of the source code. Function and API names are retained as supplied.

```
#define TA_FAIL    0x01
#define TA_SUCCESS 0x00

int16 Api_HandShakeFgpModule(uint32 timeov)
{
    int32 time;
    int16 result = TA_FAIL;
    uint8 cmd_buf[8], data_buf[8];

    time = timeov;
    Make_OneCmd(cmd_buf, 0xfe, 0, 0);   // Build command 0xFE in cmd_buf
    while (time > 0)                    // Exit if no response within timeout
    {
        Api_Send_Data(cmd_buf, 8, 0);  // Send command; repeat every 10 ms
        result = Api_GetCommand(data_buf, 10); // Wait up to 10 ms for reply
        if (result == TA_SUCCESS)
            break;
        time -= 10;
    }
    if (result != TA_SUCCESS)
        return -result;

    Api_Clr_Buf();                      // Clear lower-level buffer after 0xFE reply
    Make_OneCmd(cmd_buf, 0xfd, 0, 0);  // Build second command, 0xFD
    Api_Send_Data(cmd_buf, 8, 0);      // Send 0xFD command
    time = 30;
    while (time--)
    {
        result = Api_GetCommand(data_buf, 1); // Poll every 1 ms
        if (result == TA_SUCCESS &&
            data_buf[1] == 0xfd &&
            data_buf[4] == TA_FAIL)
            break;                      // Check response to command 0xFD
    }
    if (time)
        return TA_SUCCESS;              // Handshake succeeded
    else
        return TA_FAIL;                 // Handshake failed
}
```

How the handshake works

This method relies on a basic behavior: for an unsupported command, the Tuzheng module returns a TA_FAIL response sequence.

First, the host repeatedly sends the unsupported 0xFE command at 10 ms intervals. The command frame is: 0xF5 0xFE 0x00 0x00 0x00 0x00 0xFE 0xF5. If the module cannot yet respond—for example, because it has not finished powering up and initializing, is acquiring an image, or is running its algorithm—the module's lower-level serial buffer may accumulate multiple 0xFE commands. The host continues sending 0xFE until a response is received or the timeout expires.

Receiving the first 0xFE response indicates that the module has entered a valid command-response sequence. Under normal conditions, additional responses to previously buffered 0xFE commands may follow.

The host then sends exactly one different command, 0xFD, with frame: 0xF5 0xFD 0x00 0x00 0x00 0x00 0xFD 0xF5. Receiving the response to 0xFD confirms that the module has fully entered the command-waiting state.

This method can be used for the power-up initialization handshake, after leaving image acquisition, or whenever the current state of the fingerprint module is unknown.

Appendix 2: Customer Usage Notes

a. Host MCU UART usage

The host MCU software should follow this sequence: after powering on the fingerprint module's VCC, configure the MCU UART for normal operation. Before powering off the fingerprint module's VCC, configure the MCU UART pins as high-impedance inputs to prevent the host MCU from back-powering the fingerprint module. This guidance assumes that the fingerprint module's VCC should be at 0 V after power-off.

b. Protocol usage

Do not power off the fingerprint sensor while it is executing an image-acquisition command, such as enrollment or matching. If the host MCU must exit the acquisition process early, use a Tuzheng command to interrupt it; power off the sensor only after the interruption succeeds.

Example: after a finger wakes the system, the system sends the 0x0C matching command. If no finger input is detected for 3 seconds, first send an interruption command (any invalid command that follows the standard 8-byte format, such as 0xF5 0xFE 0x00 0x00 0x00 0x00 0xFE 0xF5), and then power off the sensor. Do not cut power directly without exiting the image-acquisition process.

c. Use a separate LDO to power the fingerprint sensor

If the fingerprint module's VCC shares a supply rail with the host MCU, the relatively high startup current of the fingerprint module may pull down the MCU supply voltage and cause the MCU to reset. The source recommends powering the fingerprint module through a separate LDO.

Revision History

Version	Change	Responsible person
V1.0	Initial document	Lucas